

Command & Control Frameworks

Why C2 is the C in adversary-simulation

An engagement needs a *stable* channel back to the operator: command & control. The framework is your remote shell with panache.

The mainstream C2 family

Framework	Notes
Cobalt Strike	The classic commerical; beacons, malleable C2, aggressor scripts. (Named but license-gated – most red teams buy it, students learn with free)
Sliver (free/open)	Modern alternative; implants for Win/Linux/Mac; DNS/mTLS/HTTP etc.
Mythic	Free/OSS agent + plugin architecture
Metasploit (payload + meterpreter)	Classic; fine for labs
HAVOC / SilentTrinity	Enthusiast-grade, learning-fun

Beacon / implant anatomy

Implant (agent) sleeps -> wakes on jitter -> beacons to listener
-> receives commands, executes, exfiltrates via the channel
Malleable C2 profile (CS): taints traffic (JSP/HTML-like) to blend

Listener profiles & evasion

HTTP/HTTPS listeners on 80/443 (blend); DNS listeners (quiet, slow)
Jitter + sleep avoids beacon periodicity detection

Malleable profiles change the wire shape (user-agent, HTML padding) to defeat JA3/beaconing rules

'Secret guard': beacon checks for its own fingerprint-causing red flags

Defense view (so you learn the rules to beat)

Beaconing detection: fixed periodicity + same-size payloads

JA3/S fingerprinting of the TLS handshake

Process-context: beacons from weird PIDs/Logs-in-Office

Live hunting: beacon callbacks = the #1 C2 tell

OPSEC of your own C2

Split control plane from the EDR-visible plane

Never run C2 on your own corp infra; pivot box isolated

Log your actions (engagement trail) but keep the C2 logged-in-layer separate

C2 is the skeleton of an engagement. Free tools (Sliver/Mythic) teach the craft; evasion insights become detection-alert material for your blue skill.

[← Red Team vs Pentest vs Adver...](#)

[Red Team Lifecycle & OPSEC →](#)